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Cluster-based spectral models for a robust assessment of soil properties

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ABSTRACT

Soil spectroscopy has proven efficient for providing quantitative properties of soil chromophores and soil classification. However, building generic/global models might bring to light some problems which stem from the false premise that each chromophore affects all soil types equally. The objective of this paper is to offer a new cluster-based approach for the assessment of soil properties and compensating issues that might rise from using heterogeneous soil dataset. For that purpose, 1480 soil samples from varying climate zones in Israel were used. All samples were air-dried, ground, and sieved to 2 mm, then $CaCO_3$, organic matter (OM), clay and sand contents together with the spectral signature were measured according to laboratory protocols. The spectral analysis was performed using the Spectral Angle Mapper (SAM) algorithm, spectral gradient and k-means clusters to split the dataset into distinct clusters. Subsequently, cluster-based models were performed and compared with generic models. Later, we focused on the OM parameter and applied the OM detection limit approach for a more robust and accurate assessment. Our results show that soil property prediction improved significantly when using the cluster-based models compared to the generic models and therefore should be considered when dealing with large and heterogeneous databases.

1. Introduction

Soils are classified by common properties in order to acquire knowledge about the pedogenesis processes that control similarities within a group and dissimilarities among groups (Birkeland, 1999). Two of the most common classification systems for soils are the United States Department of Agriculture (USDA) soil taxonomy (Soil Survey Staff, 2014) and the World Reference Base for Soil Resources (WRB) (IUSS Working Group, 2015) which use a hierarchical (level-based) approach in order to form and refine the soil classes. General soil surveys attempt to categorize soil into different types by using a similar collection of physical, chemical and biological properties and are classified by the diagnostic horizons properties within the soil profile (Ashman and Puri, 2002).

The spectral-based remote sensing is a new technology which can be used from various domains (e.g. laboratory, field, air and space borne). Over the past decades, spectral information has proven to be a powerful tool to assess soil properties as well as other components (e.g. vegetation and water bodies), in order to monitor the environment. In soils the reflectance spectrum across the Visible, Near Infrared and Shortwave Infrared (VNIR–SWIR) region (400–2500 nm) is exploited for quantitative assessment of multiple important soil properties by many workers (Ackerson et al., 2015; Ben-Dor, 2002; Ben-Dor and Banin, 1995; Chabrillat et al., 2013; Gomez et al., 2008; Ogen et al., 2017; Rossel et al., 2009; Zornoza et al., 2008). The spectral information which originates from the interaction between electromagnetic (EM) waves and matter, enables precise identifications of soil properties without the need to rely on wet laboratory analysis. Chromophores are defined as soil properties (chemical or physical) that affect the shape and nature of a soil spectrum that are resulted in from the EM and matter interaction (Ben-Dor et al., 2008). The chemical chromophores include minerals (mostly clay minerals, iron oxides, primary minerals – feldspar, salt, carbonates, phosphates), organic matter (OM) and water, while the physical chromophores are mostly related to particle size distribution (Ben-Dor et al., 1999, 2008; Clark, 1999).

The spectral-based approach is used either for providing quantitative assessment about the soil chromophores (Chabrillat et al., 2013; Conforti et al., 2017; Jiang et al., 2016; Madhavan et al., 2016; Nawar et al., 2016; O'Rourke et al., 2016; Rossel and Webster, 2012; Sorenson et al., 2017) or for the purpose of soil classification (McBratney and de Gruijter, 1992; Odgers et al., 2011; Rossiter et al., 2017; Teng et al., 2018; Vasques et al., 2014; Rossel and Webster, 2011), but mostly not simultaneously for both. However, the link between the optical data and soil classification is still indistinct. The reason for this may stem

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from the complexity of the soil classification, particularly in higher levels of classification due to: i) Integration with non-optical factors such as climate, geographic location, topography and human activity, which are not organized hierarchically in the classification system. For example, in the USDA soil taxonomy, the climate factor can be used either in the Order level (e.g. Aridisol), Suborder level (e.g. Xeralf), or Great Group (e.g. Xerofluvent). ii) Identification of diagnostic horizons, which demands high repetitive spectral measurements along the profile.

An attempt to exploit soil spectral proximate analysis and soil classification had been conducted by Zeng et al. (2017), where soils were classified based on the spectral prediction of several physio-chemical properties. The overall allocation accuracy at Order, Suborder, Group and Subgroup level was 98.5%, 98.5%, 87.7% and 76.0%, respectively. Another important study regarding this issue was performed by Pinheiro et al. (2017) in the Amazon basin. Their results indicate that soil property model performances were affected by landscape position, soil order and soil depth. In other words, the outliers in the prediction models were caused by the heterogeneity of the samples. Therefore, they concluded that in order to improve the efficiency of the model, it is recommended to produce more homogeneous groups, beforehand. Vasques et al. (2010) estimated soil organic carbon using models which are based on two criteria: (i) mineral versus organic horizons and (ii) soil orders (USDA soil taxonomy). The global model resulted in $R^2 = 0.74$ with RMSEP = 0.76%, and the mineral and organic horizons models were $R^2 = 0.71$ with RMSEP = 0.65% and $R^2 = 0.39$ with RMSEP = 0.33%, respectively. Results for the soil order models ranged from $R^2 = 0.53$ to 0.82 with RMSEP = 0.92 to 0.57%. They concluded that building models for each soil order has the potential to improve soil organic carbon prediction in mineral soils, however, the drawback of the method is that soil taxonomy information is not always available.

Another classic method used in order to increase the accuracy of soil prediction models is by spiking the calibration dataset with samples taken from a target site (spiked samples) which were analyzed in the laboratory (Brown, 2007; Guerrero et al., 2010, 2016; Shepherd and Walsh, 2002). The study concluded that the higher the proportion of the spiking samples, the higher the accuracy. These studies share the same conclusions as Sankey et al. (2008) who found that a global spectral library augmented with a local dataset improves the prediction of clay, organic carbon and inorganic carbon.

In many studies (e.g. Brown, 2007; Brown et al., 2006; Rossel, 2009; Shepherd and Walsh, 2002), the main focus is on developing soil property models using large soil databases. However, building generic/global models might bring to light some problems which stem from the premise that each chromophore affects all soil types equally, a claim which has been proven false in Ogen et al. (2018). In addition, building generic soil property models based on the soil type is incorrect because it is a subjective and complex method, and requires skilled personnel for the description of diagnostic horizon. Araújo et al. (2014) and later Shi et al. (2014), showed that a preliminary division of the dataset into subsets or clusters, may improve the prediction performance of soil organic matter and clay contents by reducing the root mean square error (RMSE). Their results were supported by a more recent study conducted by Castaldi et al. (2018), who concluded that an initial clustering of a large database (in that case, the European LUCAS topsoil database) may result in a good prediction of organic carbon (OC) in arable soils. Furthermore, Ogen et al. (2018) performed an assessment of OM and found that soil types differed in their soil spectral detection limit (SSDL) values. They concluded that samples with predicted OM values below a certain threshold have limited reliability and recommended a prior classification of the data for providing accurate models.

The objective of this study is to assess the performance of cluster-based models in a comparison with a generic model for $CaCO_3$, OM, Clay and Sand contents. For this purpose, the Spectral Angle Mapper (SAM) and the spectral gradient (m) parameters have been used as classifiers based on Ogen et al. (2017). SAM can range from 0 to 1 and the lower the value, the greater the similarity between the two spectra, regardless of their albedo. For calculating the spectral gradient we presume that the typical soil spectrum is relatively low at 500 nm and increases towards the red and NIR region (Van der Meer and de Jong, 2011). High gradient values indicate that a measured sample reflects more light, hence, it is brighter, due to its chemical-physical components.

We anticipate that the cluster-based approach will solve the problems that arose from large-sized models and soil type-based modeling and will reduce the need for spiking. In addition, we assume that this method could be applicable to any given soil sample which would be spectrally classified, and more importantly, results in a more robust soil properties assessment.

2. Materials and methods

2.1. The soil database

The database consists of 1480 soil samples which were collected by Field Service Laboratories and Agricultural Research Organization, Volcani Center, for the purpose of soil properties assessment and soil classification. The sampling was conducted in various geographic areas in Israel at varied depths, which covers the coastal plains, lowlands, valleys and mountain terrains in the Csa, BSh, BWh climate zones according to Köppen classification (Kottek et al., 2006).

All samples were air-dried, ground and sieved to ≤ 2 mm. The soil properties included clay and sand contents measurements using hydrometer (Bouyoucos, 1936), $CaCO_3$ was measured using a Calcimeter and OM was determined by the wet combustion method (Walkley and Black, 1934) in which the organic carbon in the soil sample is measured by oxidation. Then, based on the assumption that OM contains 58% organic carbon, a conversion factor of 1.724 is used in order to calculate OM (%). Only samples that contain data of at least one of these properties (CaCO3, OM, Clay or Sand contents), were brought to the Remote Sensing Laboratory at Tel-Aviv University for spectral measurement as a part of a national project for building a soil spectral library (funded by the Ministry of Agriculture).

The distribution of samples for each parameter is given by the violin and box plots in Fig. 1, where the number of samples for each measure is shown on top of each plot. The gray square includes all samples in the second and third quartiles with the black line showing the median and the violin plot around the box plot show the kernel probability density of the data at different values (Hintze and Nelson, 1998).

2.2. Laboratory spectral measurements

Soil samples were measured with an Analytical Spectral Devices Pro FR (ASD Inc., Boulder, USA) spectrometer using a contact probe equipped with a halogen bulb and Spectralon® as a white reference. The spectrometer has three detectors with 1 nm spectral resolution that supply 2151 bands in VNIR (350–1000 nm), SWIR1 (1000–1790 nm) and SWIR2 (1790–2500 nm). Samples were measured using the Internal Soil Standard (ISS) method proposed by Ben-Dor et al. (2015). The noisy bands at the end of the SWIR2 detector were removed leaving the spectral range between 400 and 2450 nm for spectral analyses.

2.3. Spectral analysis

The SAM and the spectral gradient (m) were calculated. SAM algorithm, determines the similarity between two spectra by treating them as vectors in a space with dimensionality equal to the number of bands and calculating the angle (θ) between them (Kruse et al., 1993). We used a unity reference spectrum with reflectance values equals to 1 for all wavelengths as a reference. The SAM algorithm is given by Eq. (1):

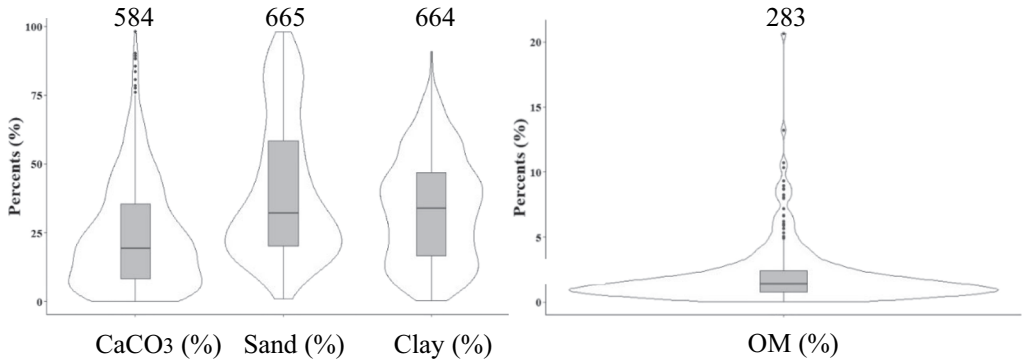


Fig. 1. Violin and box plots of soil samples for $CaCO_3$, sand, clay and OM. The number of samples for each measure is shown on top of each plot.

$$SAM = cos(Rr, Rm) \quad (1)$$

where *Rr* and *Rm* are the reference spectrum and the measured spectrum, respectively. The spectral gradient between 500 and 800 nm was calculated as was suggested by Ogen et al. (2017) and is given by Eq. (2):

$$\text{Gradient}\,(m) = \frac{Ref_{800} - Ref_{500}}{\lambda_{800} - \lambda_{500}} * 1000 \quad (2)$$

where *Ref* is the reflectance value in a given wavelength and λ is the wavelength in nm.

However, since the dataset used in this study contains numerous soil samples with high spectral variance, the SAM and gradient values were standardized to Z-score.

2.4. Spectral clustering

For clustering and classification purposes, we used the method suggested by Ogen et al. (2017) where the SAM and gradient parameters were calculated for each sample. The number of k-centroids was selected using the Agglomeration schedule method (the 'elbow' method) in SPSS statistical software (IBM Corp., 2011). Subsequently, the centroids were randomly generated in the SAM-Gradient space in order to define the desired number of clusters followed by an assignment of all other samples to their nearest centroid using the calculation of Euclidean distance (Forgy, 1965).

2.5. Soil properties modeling and assessment

Prediction models were built in two methods; generic models and cluster-based models. The generic models use the entire soil database in order to build models for each property using Partial Least Squares (PLS) regression models. The cluster-based models first perform initial classification using k-means as suggested by Ogen et al. (2017) followed by building a prediction model for each cluster. The purpose of the initial classification is to group all samples that exhibit similar spectral features into clusters in order to build models for each of them, using the PLS_Toolbox (2016) (Eigenvector Research, Inc., Manson, WA USA 98831; http://www.eigenvector.com). Ogen et al. (2017) performed their study at a field scale and therefore used the unstandardized SAM and gradient values. However, the database of this study is characterized by a high variability of the soil samples and therefore, we used the standardized (Z score) values of SAM and gradient before k-means was applied.

We used the following preprocessing methods which are implemented in PLS_Toolbox onto the spectral data: smoothing (Savitzky and Golay, 1964), baseline (PLS_Toolbox, 2016), first derivative and second derivative (Savitzky and Golay, 1964) or a combination. These combinations were then applied either on the entire spectral range (for $CaCO_3$, Clay and Sand contents) or on the 500–1000 nm spectral range which is sensitive to OM contents (Cozzolino and Morón, 2006; Ogen et al., 2018; Tian et al., 2013). Then, the dataset was split into calibration (80%) and validation (20%) groups using either the Kennard-Stone method (Kennard and Stone, 1969) or the Onion method (PLS_Toolbox, 2016) and after all options were examined for both the generic and the cluster-based models, the best fit model was selected using RMSE and R^2 values.

However, using only R^2 and RMSE in this case is not sufficient. While R^2 studies the relationship between two variables, it does not quantify and assess the differences between the OM measurement methods. For that purpose, the Bland-Altman method was used, which is the quantification of the agreement between two quantitative measurements by studying the mean differences (bias) and constructing limits of agreement (LoA) between them (Altman and Bland, 1983; Eksborg, 1981). The bias is calculated using Eq. (3) whereas the LoA are defined using Eq. (4), if the differences are normally distributed.

$$bias = \frac{1}{n}\sum_{n=1}^{n}(A - B) \quad (3)$$

$$bias - 1.96sd\,\langle \text{LoA} \rangle\, bias + 1.96sd \quad (4)$$

where *A* is the OM (%) measured using the wet chemistry method and *B* is the OM (%) values as predicted by the spectral method. The normal distribution assumption was evaluated with the Shapiro-Wilk normality test (Shapiro and Wilk, 1965) using the SPSS software, for the hypothesis that the distribution of the observations in the dataset is normal (if sig. $<$ 0.05 normality is rejected).

Ogen et al. (2018), highlighted two important aspects when dealing with OM. (i) different soil types exhibit different interactions of the EM with OM where the dominant predictor for the soil spectral detection limit (SSDL) is the amount of OM existing in the soil. The more OM in the soil, the higher the SSDL which can be detected by spectroscopy. The lowest SSDL was found to be in the sand sample (0.35%) which had 0.27% of initial OM content. (ii) The uncertainty of the predicted value was somewhat equal in all soils (an average of ± 24% from the initial OM content).

The SSDL method uses the first derivative spectrum in order to identify and calculate the ratio between the VIS and NIR (the exact wavelengths differ from one soil to another) for a certain soil dataset and a predefined blank sample. For that purpose, the blank sample is measured 20 times and the mean and standard deviation can be calculated. Subsequently, the detection limit is determined once the ratio of a soil exceeds the mean ratio value ± 3 standard deviation of the blank sample. Moreover, any sample with predicted OM value between −3 and +3 standard deviations of the blank (± 24% on average) is considered to have a limited liability because it falls in the uncertainty range of the spectral prediction models. A more detailed description is provided in Ogen et al. (2018).

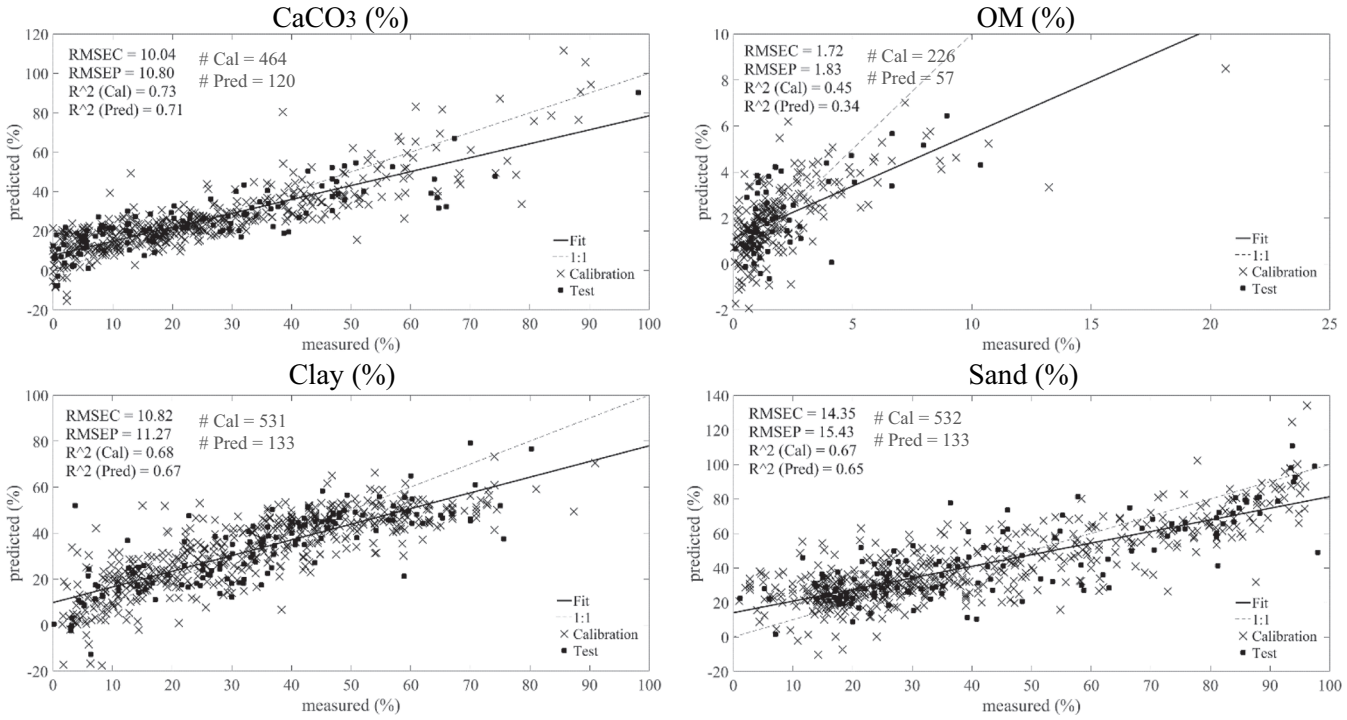


Fig. 2. Generic PLS prediction models of $CaCO_3$, OM, sand and clay.

3. Results and discussion

3.1. Soil properties prediction using generic models

Soil property prediction models of $CaCO_3$, OM, clay and sand contents were calibrated using all samples in the dataset for each property (see Fig. 1). The results are presented in Fig. 2 where all models performed well except that of OM content. The $CaCO_3$ prediction model provides the best results with RMSEP = 10.80% and R^2 = 0.71. The sand prediction model supplies RMSEP = 11.27% and R^2 = 0.67 and the clay content prediction model parameters were RMSEP = 15.43% and R^2 = 0.65. The result of the OM prediction model was unsatisfactory with RMSEP = 1.83% and R^2 = 0.34. The low performance of the OM model may stem from the heterogeneity of OM compounds and their decomposition stage (Ben-Dor et al., 1997), or due to the heterogeneity of the soils in the database (Ogen et al., 2018).

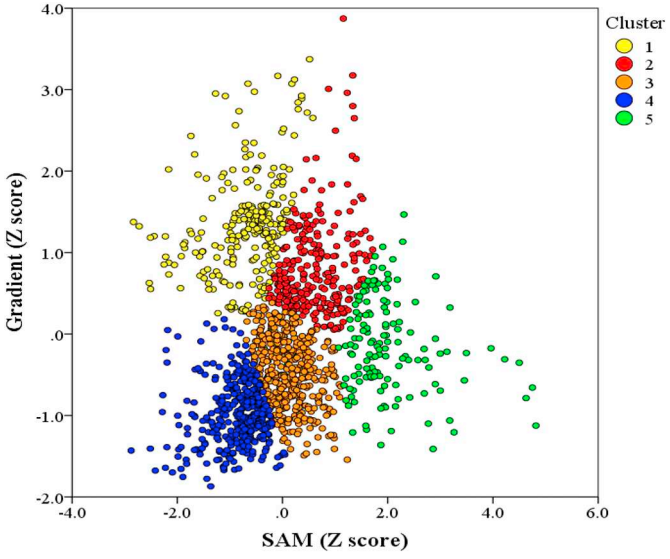


Fig. 3. The SAM-Gradient space of 1480 soil samples which were grouped into 5 clusters using k-means algorithm.

3.2. Cluster-based prediction models

The cluster-based approach projects each sample on the SAM-Gradient scatter plot and uses the k-means classifier. In order to determine the number of clusters, the largest number of clusters was selected, where each cluster contained enough samples for statistical modeling. In our case, five clusters were sufficient to be spectrally varied and still contain enough samples for analysis. Fig. 3 presents the classified samples on the SAM-Gradient plot. Subsequently, prediction models have been constructed for each cluster separately. As a result, samples in the same soil profile (and hence same sub-order) may be associated to one or more clusters and will be subject to a different model which better fits their spectral characteristics.

When applying the values of the soil properties on the SAM-Gradient space, some trends are noticeable as seen in Fig. 4. Cluster 1 contains most of the high $CaCO_3$ samples whereas the lower values are located in a half circle shape stretching between through clusters 2, 5, 3, and 4. OM does not exhibit any specific trend, although the higher amounts (> 2.5%) are present mostly in clusters 3 and 5. Sand and clay contents show opposite trends where high values of the sand are located mostly in cluster 2, but also in some regions of clusters 1 and 5, while the high clay is located mostly in clusters 3 and 4.

Because climate is the main factor for soil formation in many classification schemes, the annual precipitation (in mm) of the sampling locations is also plotted on the dataset. The distribution of the annual precipitation, corresponds to the k-means clusters (Fig. 5) and moreover, it corresponds with the distribution of soil property contents shown in Fig. 4. For example, high $CaCO_3$ content soils are found mostly in arid and semi-arid areas (< 400 mm) and most of the samples taken from the Mediterranean climate region (> 400 mm) are located in clusters 3–5.

Our method suggests that any given sample should be modeled only after it has been classified based on its spectral response (SAM-Gradient plot) and not on the soil type of the entire profile. An example is given

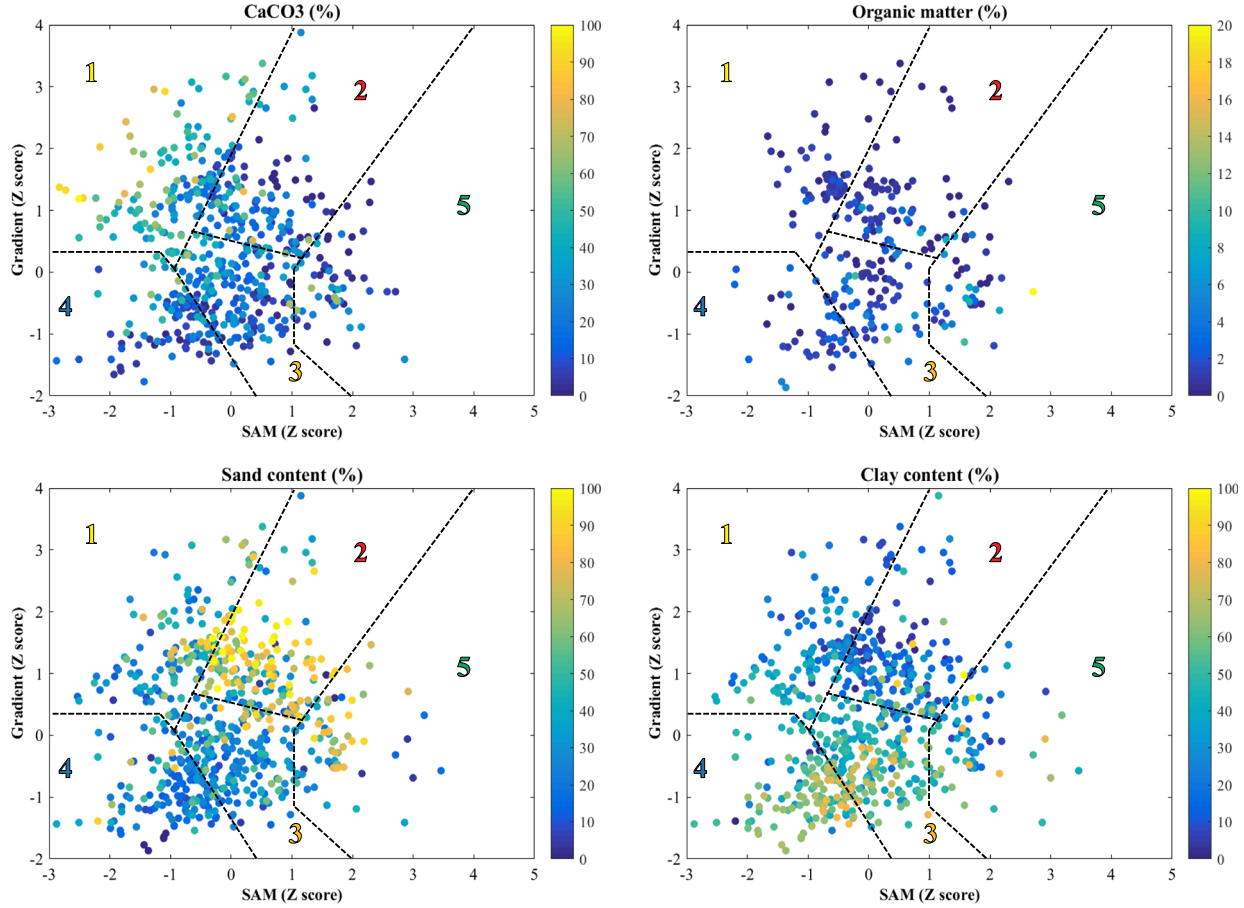


Fig. 4. The distributions of soil properties values in the database. The dashed line marks the border between the k-means clusters.

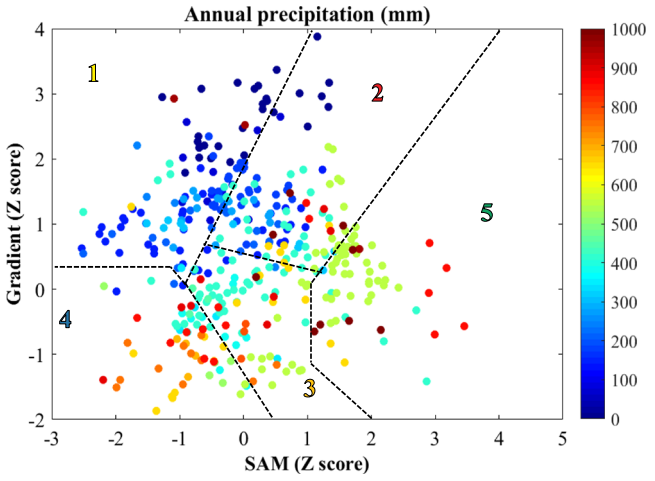


Fig. 5. The distributions of the annual precipitation (mm) in the location of the sampling sites. The dashed line marks the border between the k-means clusters.

in Fig. 6 by a Terra Rosa soil profile (Chromic Luvisol in the WRB classification) with an annual precipitation of 650 mm.

In general, Terra Rosa soils are developed in the Mediterranean climate region on Limestone or Dolomite (Singer, 2007). During pedogenesis, the calcium/magnesium carbonate is washed away. However, an accumulation of a secondary calcium carbonate may occur in lower strata creating a calcic horizon. The exceptional sample in cluster 1 (Fig. 6a) is related to the calcic horizon which holds similar spectral characteristics as Calcisol taken from an arid area; High reflectance values and a significant spectral feature at 2300–2350 nm (Fig. 6b). Additionally, the samples above the calcic horizon show different spectral signature than the samples below it which is expressed mostly between 500 and 1000 nm. The difference is related to the high contents of OM in the upper horizons (10.4, 4.5 and 2.1% in the 0–20, 20–50 and 50–95 cm, respectively) and it is also expressed in the SAM parameter where the upper horizons have higher SAM values than the lower horizons (Fig. 6a). Subsequently, the soil property models were calibrated and validated for each cluster separately and the combined predicted values are shown in Fig. 7.

It was found that a significant improvement has been obtained using the cluster-based models compared to the generic models. $CaCO_3$ prediction results were improved from $R^2 = 0.71$ in the generic models to values between 0.82 and 0.93 in the cluster-based along with lower values of RMSEP. The OM models exhibit the most significant improvement from $R^2 = 0.34$ to 0.72–0.81 in all clusters. However, the RMSEP was increased only in cluster 5 from 1.83 to 2.31. It should be noted while producing the models, a total of five outliers were detected using the Q residuals (the sum of squares of each sample in the error matrix), and were removed from the models. The clay content models were improved from $R^2 = 0.67$ to 0.71–0.78, except for cluster 4 which was reduced to $R^2 = 0.63$. The RMSEP was reduced significantly in all clusters, from 11.27 to values below 7.54, except for cluster 5 where it was slightly increased to 13.4. The reason may stem from the high variance of clay contents and a small number of samples. The sand generic model gave $R^2 = 0.65$ and RMSEP = 15.43 and the cluster-based models resulted in significant improvement of R^2 in clusters 4 ($R^2 = 0.81$) and 5 ($R^2 = 0.88$), satisfactory improvement in clusters 1 ($R^2 = 0.70$) and cluster 2 ($R^2 = 0.76$), while in cluster 3 it showed no

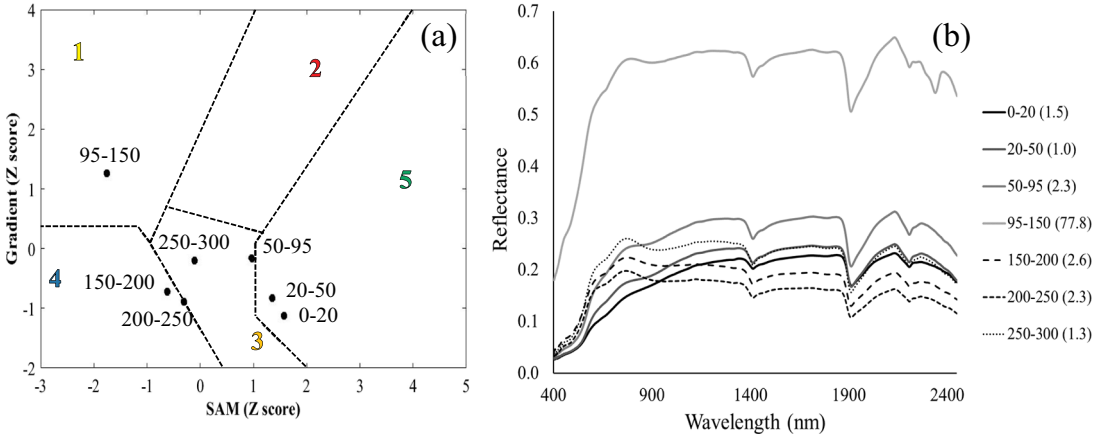


Fig. 6. A plot of 7 samples taken from a Terra Rosa profile. All samples are grouped in clusters 3–5 (except for the sample in 95–150 cm which is associated with cluster 1 (a)). The spectrum of each sample taken from a different depth (cm) with the percentage of $CaCO_3$ (%) in brackets (b).

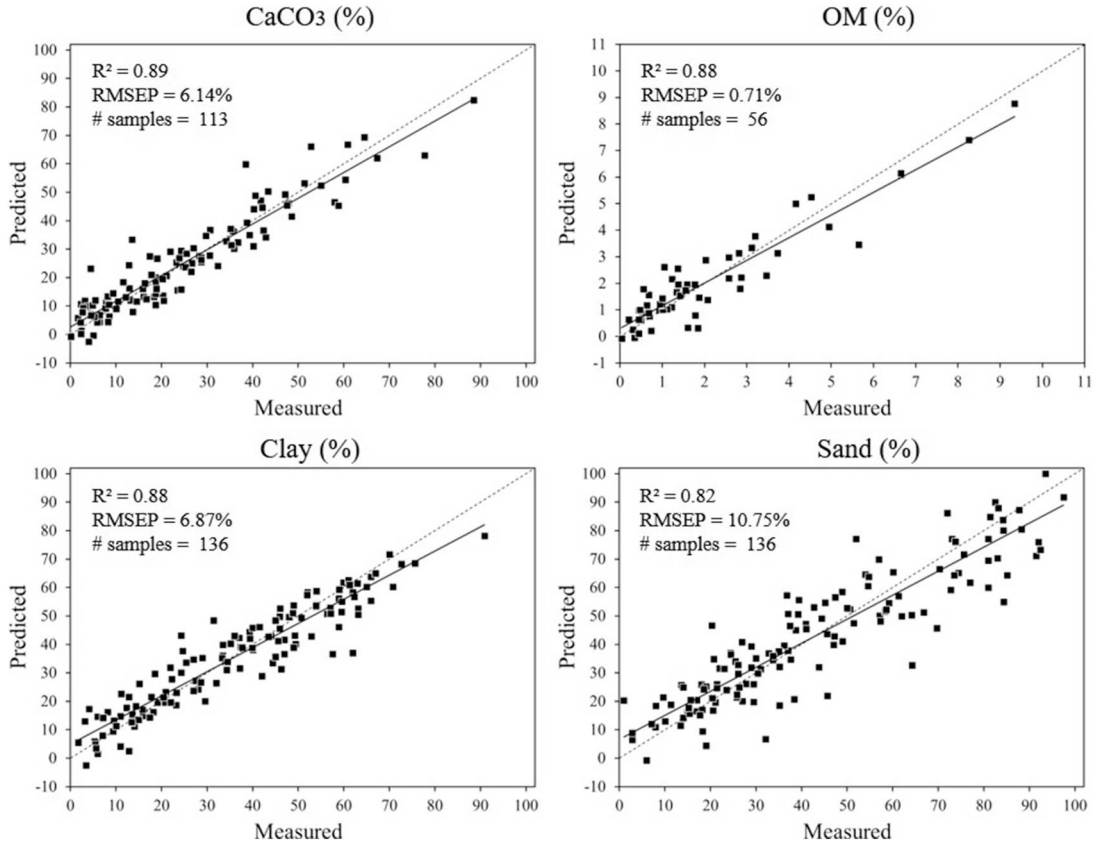


Fig. 7. The combined prediction results of clusters 1–5 for each soil property.

Table 1
Summary of the RMSEP and the linear coefficient of determination (R^2) values for the generic model in comparison with the cluster-based models and the combined dataset.

		Generic	Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Combined
$CaCO_3$	RMSEP (%)	10.8	8.2	5.92	4.02	4.42	5.55	6.14
	R^2	0.71	0.84	0.82	0.84	0.93	0.92	0.89
OM	RMSEP (%)	1.83	0.29	1.22	0.96	0.59	2.31	0.71
	R^2	0.34	0.81	0.78	0.72	0.73	0.76	0.88
Clay	RMSEP (%)	11.27	6.51	7.47	7.49	7.54	13.04	6.87
	R^2	0.67	0.75	0.78	0.71	0.63	0.76	0.88
Sand	RMSEP (%)	15.43	13.45	12.15	10.41	9.11	9.34	10.75
	R^2	0.65	0.70	0.76	0.65	0.81	0.88	0.82

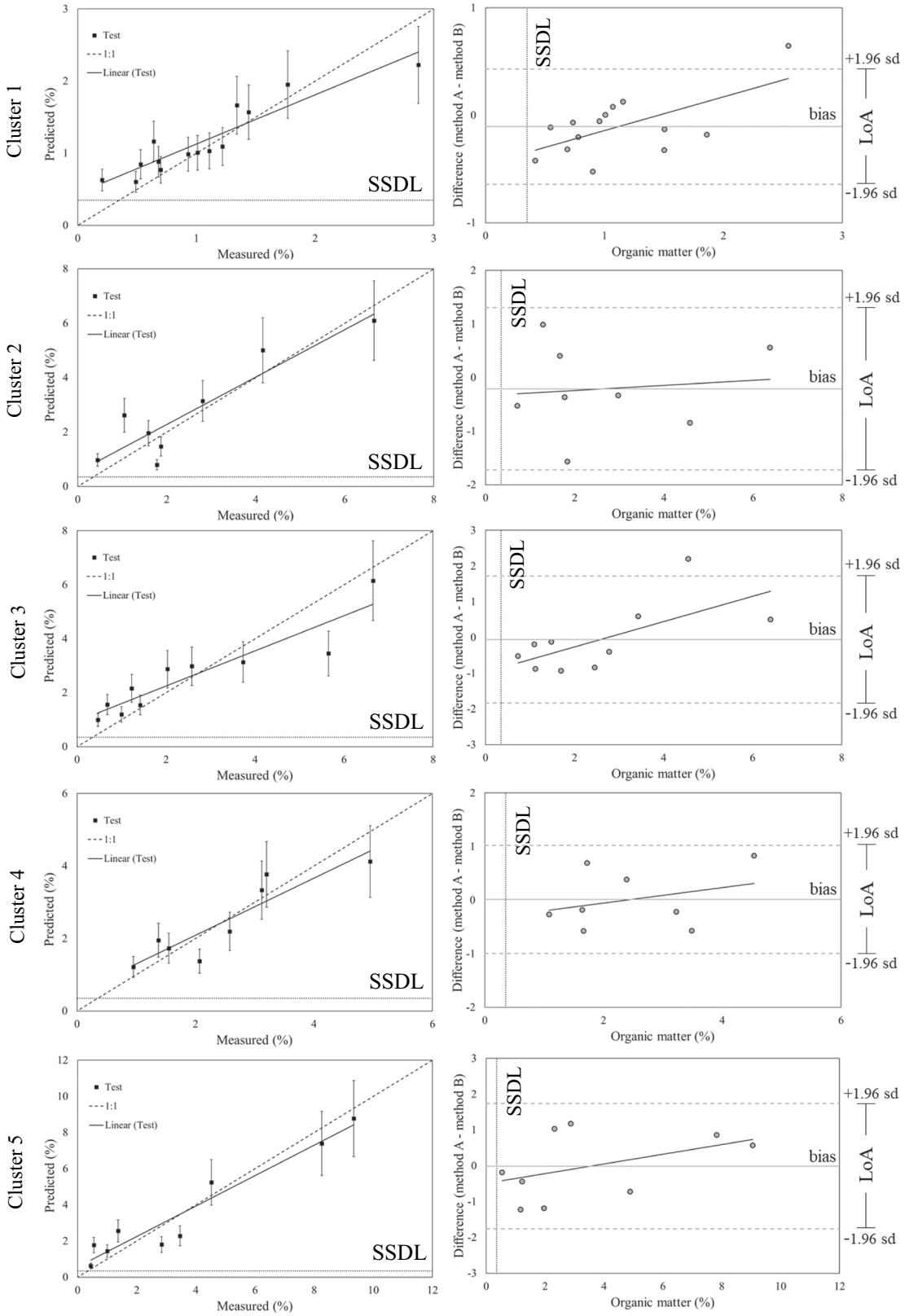


Fig. 8. The measured vs. predicted of OM contents in clusters 1–5 after removing the test samples that contain predicted OM lower than the SSDL. Error bars were added to show the ± 24% uncertainty range of the predicted value. The right graphs show the quantification of the agreement between the measured and predicted values of OM, using the Bland-Altman method.

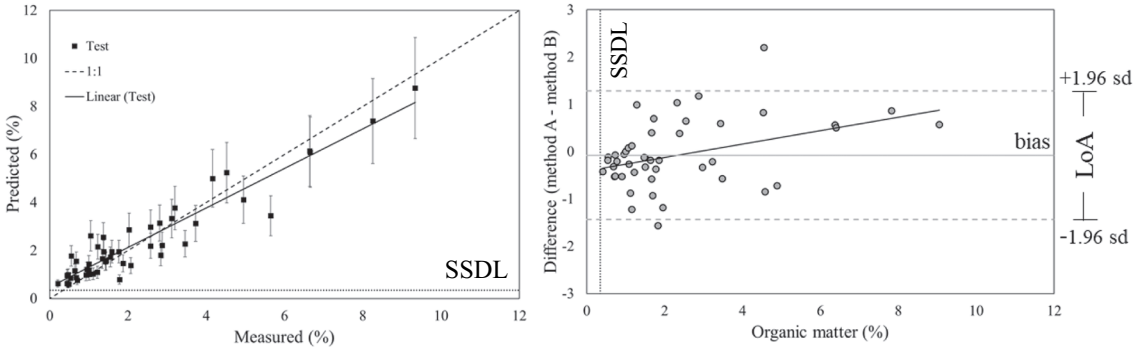


Fig. 9. The measured vs. predicted OM contents in the combined dataset with the error bars of ± 24% uncertainty range of the predicted value. The Bland-Altman method is shown on the right.

Table 2
Summary of the RMSEP and the linear coefficient of determination (R^2), the bias and standard deviation (sd) of the Bland-Altman method and the sig. value of the Shapiro-Wilk normality test for each cluster and the combined data.

		Cluster 1	Cluster 2	Cluster 3	Cluster 4	Cluster 5	Combined
No. of test samples		14	8	10	8	9	49
OM	RMSEP (%)	0.29	0.80	0.91	0.52	0.89	0.69
	R^2	0.85	0.83	0.84	0.82	0.93	0.89
Bland-Altman	Bias	−0.11	−0.21	−0.06	0.01	−0.01	−0.08
	sd	0.27	0.77	0.91	0.52	0.89	0.69
Shapiro-Wilk	sig.	0.11	0.12	0.16	0.42	0.07	0.00

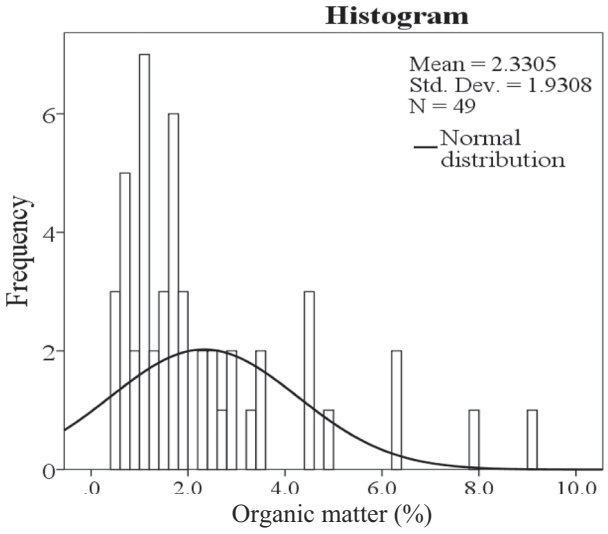


Fig. 10. Histogram of OM in the combined dataset.

change from the generic model. However, RMSEP parameter decreased in all clusters. Subsequently, the predicted values in all clusters were combined for each soil property. The combined test results show an improvement for the prediction of the soil properties with an overall R^2 between 0.82 and 0.89. Results are summarized in Table 1.

3.3. Applying the OM detection limit on the models

Using the method developed by Ogen et al. (2018), the soil spectral detection limit along with the uncertainty range were calculated and applied on the predicted values of OM and are presented in the left column of Fig. 8. Any sample which was predicted to contain < 0.35% of OM is considered to be a wrong assessment and therefore were eliminated from the prediction model. The scatter plots shown on the right column of Fig. 8 are the Bland-Altman method in which the difference between the two OM measurements in each cluster is plotted against their mean value. Fig. 9 presents the measured vs. predicted and the Bland-Altman scatter plot for the combined dataset where we would expect that 95% of the differences will be located in the LoA range, where the *bias* ± 1.96*sd*, following the normal distribution requirement. Furthermore, the results of all samples were combined into one dataset and the measured vs. predicted along with the Bland-Altman are presented in Fig. 8. The overall results are given in Table 2 for the RMSEP and R^2 values of OM for each separate cluster and for the combined dataset. Additionally, the bias and the standard deviation (sd) values of the Bland-Altman method are presented together with the Shapiro-Wilk normality test.

After applying the SSDL on the dataset, the RMSEP values decreased for all clusters (except for cluster 1 where the value remains 0.29), whereas the R^2 value increased (Table 2). The combined data delivered excellent results where the RMSEP and R^2 values are 0.69 and 0.89, respectively. Subsequently, the agreement between the two quantitative measures were tested in which the negative bias means that on average, method B (spectral) measures higher OM values than method A (wet chemistry). The Shapiro-Wilk normality test shows that the Bland-Altman results are valid (sig. > 0.05) for all clusters except for the combined dataset (sig. = 0.00). The reason for this lies in the skewed distribution of the samples to low OM values as presented in Fig. 10.

In order to check whether the Bland-Altman method can still be valid despite the non-normal distribution of samples in the combined data, it was divided into two subgroups. The division is based on the 2% OM content, which was found to be the threshold in the OM chemometric approach (Ben-Dor et al., 1997; Ogen et al., 2018). The low OM content (< 2%) subgroup contains 31 samples and the high OM content (> 2%) subgroup contains 18 samples. Fig. 11a and b present the samples distribution and the Bland-Altman results for the < 2% OM and Fig. 11c and d present the samples distribution and the Bland-Altman results for the > 2% OM. Results are summarized in Table 3.

Due to the skewed distribution of the OM, the Bland-Altman method cannot be applied on the combined dataset. However, under the 2%

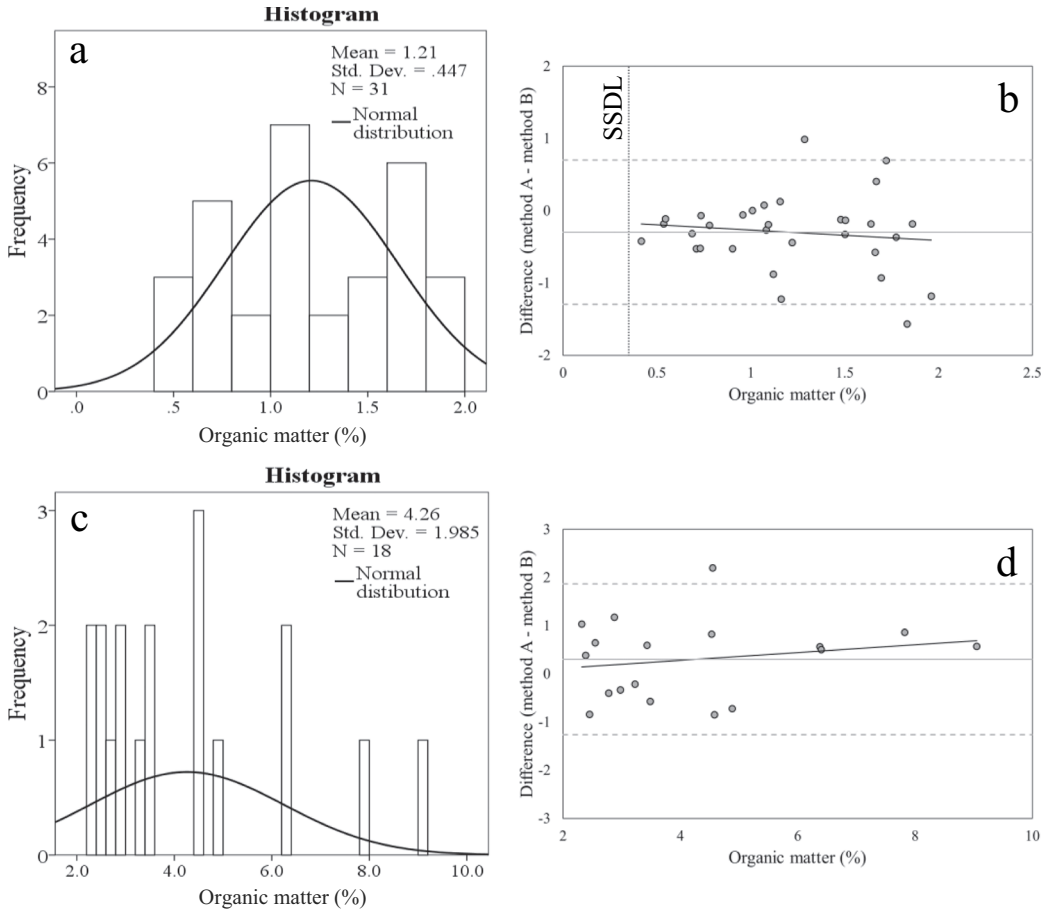


Fig. 11. (a) The distribution of samples and (b) the Bland-Altman method of the low OM content (< 2.0%) and (c) the distribution of samples and (d) the Bland-Altman method of the high OM content (> 2.0%).

Table 3
Summary of the bias and standard deviation (sd) of the Bland-Altman method and the sig. value of the Shapiro-Wilk normality test for the < 2% OM samples and the > 2% OM samples from the combined dataset.

		< 2% OM	> 2% OM
No. of test samples		31	18
Bland-Altman	Bias	−0.3	0.3
	sd	0.51	0.80
Shapiro-Wilk	sig.	0.18	0.01

threshold, the Shapiro-Wilk test gives sig. value of 0.18, meaning that OM content is normally distributed in contrast to the sig. value of 0.01 above 2% (Table 3).

4. Summary and conclusions

In order to build reliable soil proxy models, recent studies have suggested to group the samples into more homogeneous manner (Ogen et al., 2017, 2018; Pinheiro et al., 2017). However, although prediction models based on soil types or diagnostic horizons may provide some improvement, the taxonomic information is not always available (Vasques et al., 2010).

We suggest a cluster-based approach in order to build more reliable and precise soil property models. While the generic approach provides a lack of accuracy due to the heterogeneous nature of the soils in question, building spectral cluster-based models provided a significant improvement in accuracy. The cluster-based approach has several advantages: (i) It clusters soil samples according their spectral information which is a more correct comparative approach for constructing spectral models, (ii) it does not consider the soil profile as one entity but rather sees each soil sample as a separate entity upon which the right model will be operated, as shown in Fig. 6, and (iii) it does not require knowledge of the soil type or to use sample spiking. Furthermore, including the SSDL method not only improved the OM models, but also provided the range of uncertainty of the predicted values.

The main conclusions for this work are:

1. The soil properties generic models supply good results for the $CaCO_3$, clay and sand contents with R^2 = 0.71, 0.67 and 0.65, respectively. However, the OM prediction model showed low performance with R^2 = 0.34 due to the heterogeneity of OM compounds and their decomposition stage.
2. The cluster-based spectral models, provided a robust assessment of all soil properties in question with R^2 between 0.82 and 0.89 for the combined clusters results.
3. Applying the SSDL to the predicted OM data, resulted in further improvement for all clusters with R^2 between 0.82 and 0.93 and with R^2 = 0.89 for the combined data.

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Authors contributions

Y.O. collected the soil samples, organized the soil database, performed the statistical analysis and is the main author of this article. J. Z. and N. F. cataloged the samples, performed the spectral measurements and corrections. N.G. and E.B.-D. were the project supervisors.

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